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Avaliação: 6

- Verificando as máquinas

```
root@ubuntu:~$ # Em ambos os terminais, verificar o IP
ip addr show | grep inet
hostname -I
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host noprefixroute
    inet 172.30.1.2/24 brd 172.30.1.255 scope global dynamic noprefixr
oute enp1s0
    inet6 fe80::c06a:7071:bb4:7394/64 scope link
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
172.30.1.2 172.17.0.1
root@ubuntu:~$
```

- Instalando UFW (em ambas as máquinas)

```
172.30.1.2 172.17.0.1
root@ubuntu:~$ # Atualizar pacotes
sudo apt update

# Instalar UFW
sudo apt install ufw -y

# Verificar versão instalada
sudo ufw --version
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
11 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
ufw set to manually installed.
The following package was automatically installed and is no longer req
uired:
  squashfs-tools
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 11 not upgraded.
ufw 0.36.2
Copyright 2008-2023 Canonical Ltd.
root@ubuntu:~$
```

- Verificando o status inicial

```
ufw 0.36.2
Copyright 2008-2023 Canonical Ltd.
root@ubuntu:~$ # Verificar status (deve estar inactive)
sudo ufw status verbose
Status: inactive
root@ubuntu:~$
root@ubuntu:~$
```

- Configurando políticas padrão

```
sudo ufw default deny incoming

# Definir política padrão: PERMITIR tudo que sai
sudo ufw default allow outgoing
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
root@ubuntu:~$
```

- Permissão da porta SSH

```
(be sure to update your rules accordingly)
root@ubuntu:~$ # Permitir SSH na porta 22
sudo ufw allow 22/tcp

# OU usar o nome do serviço
sudo ufw allow ssh
Rules updated
Rules updated (v6)
Skipping adding existing rule
Skipping adding existing rule (v6)
root@ubuntu:~$
```

- Firewall ativado com sucesso

```
root@ubuntu:~$ # Ativar UFW
sudo ufw enable

# Confirmar com 'y' quando solicitado

# Verificar status ativo
sudo ufw status verbose
Firewall is active and enabled on system startup
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)

root@ubuntu:~$
```

- Testes de bloqueio (Máquina A: Servidor com firewall e Máquina B: Cliente que vai tentar acessar)

```
root@ubuntu:~$ # Em ambas as máquinas, instalar ferramentas de teste
sudo apt install netcat-openbsd curl telnet -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
netcat-openbsd is already the newest version (1.226-1ubuntu2).
netcat-openbsd set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.9).
telnet is already the newest version (0.17+2.5-3ubuntu4.1).
telnet set to manually installed.
The following package was automatically installed and is no longer required:
  squashfs-tools
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 11 not upgraded.
```

- Verificar SSH (Porta 22) Na Máquina A (Servidor com firewall):

```
May 27 00:22:36 ubuntu sudo[3178]: pam_unix(sudo:session): session op>
May 27 00:22:37 ubuntu sudo[3178]: pam_unix(sudo:session): session cl>
May 27 00:23:06 ubuntu sudo[3188]: root : TTY=pts/3 ; PWD=/root ;>
May 27 00:23:06 ubuntu sudo[3188]: pam_unix(sudo:session): session op>
May 27 00:23:06 ubuntu sudo[3188]: pam_unix(sudo:session): session cl>
May 27 00:23:06 ubuntu sudo[3198]: root : TTY=pts/3 ; PWD=/root ;>
May 27 00:23:06 ubuntu sudo[3198]: pam_unix(sudo:session): session op>
May 27 00:23:06 ubuntu sudo[3198]: pam_unix(sudo:session): session cl>
May 27 00:23:06 ubuntu sudo[3202]: root : TTY=pts/3 ; PWD=/root ;>
lines 1-37
```

- Verificar SSH (Porta 22) Na Máquina B (Cliente)

```
root@ubuntu:~$ # Substitua IP_DO_SERVIDOR pelo IP da Máquina A
ssh usuario@IP_DO_SERVIDOR
# OU testar porta sem conectar
nc -zv IP_DO_SERVIDOR 22
ssh: Could not resolve hostname ip_do_servidor: Name or service not known
nc: getaddrinfo for host "IP_DO_SERVIDOR" port 22: Name or service not known
root@ubuntu:~$
```

- Regra de bloqueio da porta 23 adicionada Na Máquina A (Servidor)

```
root@ubuntu:~$ # Bloquear porta 23 (Telnet)
sudo ufw deny 23/tcp

# Verificar regra adicionada
sudo ufw status numbered
Rule added
Rule added (v6)
Status: active
```

	To	Action	From
	--	-----	----
[1]	22/tcp	ALLOW IN	Anywhere
[2]	23/tcp	DENY IN	Anywhere
[3]	22/tcp (v6)	ALLOW IN	Anywhere (v6)
[4]	23/tcp (v6)	DENY IN	Anywhere (v6)

- Regra de bloqueio da porta 23 adicionada Na Na Máquina B (Cliente)

```
root@ubuntu:~$ # Tentar conectar na porta 23 (deve falhar)
nc -zv IP_DO_SERVIDOR 23

# Ou com telnet
telnet IP_DO_SERVIDOR 23
nc: getaddrinfo for host "IP_DO_SERVIDOR" port 23: Name or service not known
Server lookup failure: IP_DO_SERVIDOR:23, Name or service not known
root@ubuntu:~$
```

- Remoção da regra de bloqueio maquina A

```
Status: active

To          Action    From
--          -
22/tcp      ALLOW     Anywhere
23/tcp      DENY      Anywhere
22/tcp (v6) ALLOW     Anywhere (v6)
23/tcp (v6) DENY      Anywhere (v6)

root@ubuntu:~$
```

- Conexão restaurada após remoção do bloqueio

```
23/tcp (v6)      DENY      Anywhere (v6)

root@ubuntu:~$ # Testar conexão novamente - deve funcionar
nc -zv IP_DO_SERVIDOR 22
nc: getaddrinfo for host "IP_DO_SERVIDOR" port 22: Name or service not
known
root@ubuntu:~$
```